MBS-raw: Physical Optics for Ice Crystals

Manual

2026-03-22

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1 Overview

MBS-raw computes light scattering by non-spherical particles using the Physical Optics (PO) method. It traces geometric optics rays through the particle (reflections, refractions), then applies Kirchhoff diffraction to each output beam to obtain the far-field Mueller matrix.

Optimized version: ∼70× faster than original. AVX-512/AVX2 SIMD, OpenMP parallel, Sobol quasi-random orientations. Builds for Intel, AMD Zen 2 (EPYC 7H12), and AMD Zen 4 (Ryzen 7000 / EPYC Genoa).

2 Quick Start

```
# Simple run: hex column, Sobol orientations, auto theta grid
mbs_po --po --sobol 1024 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close

# Fully adaptive (auto everything)
mbs_po --po --adaptive 0.01 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close
```

3 CLI Reference

All flags are grouped by function. Required flags have no default; optional flags are marked with their defaults.

3.1 Particle Definition

3.1.1 `-p TYPE HEIGHT DIAMETER [EXTRA]`

Type	Shape	Parameters
1	Hexagonal column	H = height, D = diameter
2	Bullet	H, D (cap angle 62°, auto)
3	Bullet rosette	H, D, [cap height]
4	Droxtal	arg3 = sup parameter
10	Concave hexagonal	H, D, concavity depth
12	Hexagonal aggregate	H, D, number of elements

Symmetry: auto-detected. Hex column: $\beta_{\text{sym}} = \pi/2$, $\gamma_{\text{sym}} = \pi/3 \rightarrow 12\times$ reduction.

3.1.2 `-pf FILENAME`

Load particle geometry from file. `-p` argument provides the filename.

3.1.3 `-rs SIZE`

Resize particle loaded with `-pf` to maximal dimension SIZE (μm).

3.1.4 `-forced_convex` / `-forced_nonconvex`

Force convex or non-convex tracing algorithm.

3.2 Physical Parameters

3.2.1 `-w WAVELENGTH_UM`

Wavelength in micrometers. Examples: `-w 0.532` (green), `-w 1.064` (NIR).

3.2.2 `-ri N_REAL N_IMAG`

Complex refractive index $m = n_{\text{re}} + i\, n_{\text{im}}$.
Important: Without `-abs`, imaginary part is forced to zero.

3.2.3 `-abs`

Enable absorption. Requires `-w`. Uses `DiffractInclineAbs`.

3.2.4 `-n MAX_REFLECTIONS`

Value	Use case	Notes
5–6	Forward scattering, C_{sca}	$< 1\%$ error
12	General purpose	$< 5\%$ backscatter error
20	Backscattering	$< 3\%$ error
25	Reference calculations	

3.3 Method

- `-po` — Physical Optics (Kirchhoff diffraction). **Required for PO.**
- `-go` — Geometrical Optics (no diffraction).
- `-incoh` — Incoherent summation ($|J|^2$ per beam, then sum).
- `-karczewski` — Karczewski polarization matrix (experimental, same M_{11}).
- `-jones` — Output Jones matrices to file.
- `-all` — Compute all trajectory groups.

3.4 Orientations

3.4.1 `-sobol N` (recommended)

N Sobol quasi-random orientations (power of 2). Auto-uses particle symmetry.
Convergence: $O((\log N)^2/N)$, much faster than Monte Carlo $O(1/\sqrt{N})$.

x	C_{sca} ($< 2\%$)	$M_{11}(180°)$ ($< 5\%$)
10–50	128	512
50–200	256	1024
200–1000	512	4096
> 1000	1024	8192+

3.4.2 `-adaptive EPS`

Auto-determine orientations. Starts at 256, doubles until C_{sca} converges to relative accuracy EPS.

3.4.3 `-sym B G`

Override particle symmetry: $\beta \in [0, \pi/B]$, $\gamma \in [0, 2\pi/G]$.
Example: `-sym 6 2` $\rightarrow \beta \in [0, 30°]$, $\gamma \in [0, 180°]$ ($12\times$ reduction).

3.4.4 Other orientation modes

- `-random N_BETA N_GAMMA` — Uniform grid.
- `-montecarlo N` — Random orientations.
- `-fixed BETA GAMMA` — Single orientation (degrees).
- `-orientfile FILE` — Read (β, γ) pairs (radians) from file.
- `-b MIN MAX`, `-g MIN MAX` — Override ranges (degrees).

3.5 Scattering Grid

3.5.1 `-grid T1 T2 N_PHI N_THETA`

Uniform grid. **Use** $N_\phi \geq 48$; $N_\phi = 6$ causes 50–200% error.

3.5.2 `-auto_tgrid` (recommended)

Auto non-uniform θ grid with 5 adaptive zones:

Zone	Range	Step	Purpose
Fine	0 to 10×peak	peak/20	Forward peak
Transition	to 20°	geometric ×1.3	Smooth bridging
Medium	to 120°	0.5° (x<100) or 1°	Side scattering
Side/back	to 175°	1°	Backscattering
Near-backscatter	to 180°	0.25°	LDR

[width=]docs/figures/fig$_a uto_g rid_p oints.pdf$

Figure 1: Left: total N_θ and N_ϕ vs size parameter x. Right: zone breakdown of θ-grid points.

[width=0.85]docs/figures/fig$_a uto_g rid_s teps.pdf$

Figure 2: Step size $\Delta\theta$ vs scattering angle for different x. Small x has coarser steps (broad peak); large x needs fine steps near θ=0.

3.5.3 `-auto_phi`

Auto-select N_ϕ based on x: $N_\phi = \max(48,\ 6\lceil\sqrt{x}/1.5\rceil)$.

[width=]docs/figures/fig$_a uto_p hi.pdf$

Figure 3: Left: N_ϕ from auto_phi (blue) vs fixed values. Right: Q_{sca} error vs N_ϕ for different sizes—larger x needs more ϕ bins.

Recommended: use both `-auto_tgrid -auto_phi` together.

3.5.4 `-tgrid FILENAME`

Custom θ grid from file (degrees, one per line).

3.6 Multi-Size (`-sizefile`)

```
--sizefile sizes.txt   # one x = pi*D/lambda per line
```

Traces beams once, caches geometry, recomputes diffraction for each size. Output: `M_x10.dat`, `M_x20.dat`, etc.

3.7 Performance Options

- `-beam_cutoff EPS` — Skip negligible beams.
- `-shadow_off` — Disable shadow beam (debug).
- `-r RATIO` — Small beam restriction ratio (default 100).

3.8 Output

- `-o DIRNAME` — Output folder (default `M`).
- `-close` — Exit after calculation (**required for scripts**).
- `-log SECONDS` — Progress output interval.
- `-tr FILE` — Trajectory groups from file.
- `-gr` — Per-group output.

4 Output Files and Console Log

4.1 Output files

All output is written to the directory specified by `-o` (default `M`).

4.1.1 `DIRNAME.dat` — Mueller matrix

Space-separated, 18 columns, one header line + one row per θ bin.

```
ScAngle  2pi*dcos  M11  M12  M13  M14  M21  M22  M23  M24
                   M31  M32  M33  M34  M41  M42  M43  M44
```

- `ScAngle` — scattering angle θ in degrees
- `2pi*dcos` — solid angle weight: $2\pi\,\Delta\cos\theta$. Integration: $C_{\text{sca}} = \sum_j M_{11}(\theta_j) \times (2\pi\,\Delta\cos\theta_j)$
- `M11`... `M44` — 16 Mueller matrix elements (phi-averaged). For randomly-oriented particles: $M_{13} = M_{14} = M_{23} = M_{24} = M_{31} = M_{32} = M_{41} = M_{42} \approx 0$.

Useful derived quantities:

$$
\begin{aligned}
Q_{\text{sca}} &= C_{\text{sca}}/A_{\text{proj}} \\
g &= \frac{\sum M_{11}\cos\theta \cdot 2\pi\, d\cos\theta}{\sum M_{11} \cdot 2\pi\, d\cos\theta} \qquad \text{(asymmetry parameter)} \\
-M_{12}/M_{11} &= \text{degree of linear polarization}
\end{aligned}
$$

4.1.2 `DIRNAME\out.txt` — run summary

Contains particle parameters, refractive index, method, start/end time, total number of orientations. Written after computation completes.

```
Particle:  Refractive index: 1.310000
  Area:429.903838
Number of secondary reflections: 6
Wavelength (um): 0.532000
Method: Physical optics, Sobol quasi-random
Start of calc.: Sun Mar 22 11:22:44 2026
End of calc.  : Sun Mar 22 11:22:49 2026
Total time  : 4s
Total number of body orientations: 128
```

4.1.3 `DIRNAME_jones.dat` — Jones matrices (with `-jones`)

Written when `-jones` flag is used (fixed orientation only). 10 columns:

```
# theta(deg) phi(deg) J00_re J00_im J01_re J01_im J10_re J10_im J11_re J11_im
```

One row per (θ, ϕ) grid point. Complex Jones matrix $\mathbf{J}$ relates incident to scattered field: $\begin{pmatrix} E_\parallel \\ E_\perp \end{pmatrix}_{\text{sca}} = \mathbf{J} \begin{pmatrix} E_\parallel \\ E_\perp \end{pmatrix}_{\text{inc}}$.

4.1.4 `particle_for_check.dat` — particle geometry

Dumped by the program for verification. Contains vertex coordinates and facet definitions of the particle as used in the calculation.

4.1.5 `beam_log.dat` — beam diagnostics

Written for the first orientation (debug). Columns:

```
# beam_num  nActs  area  opticalPath  |Jones|  cx  cy  cz  nVertices
```

`nActs` = number of internal interactions, `cx,cy,cz` = beam exit direction. The shadow beam has `nActs` = large number and `cz` = -1.

4.2 Console output (stdout)

The program writes progress and results to stdout.

4.2.1 Auto mode output

```
Auto theta grid: x=83.5, peak_width=2.16 deg, 464 points
Auto phi: x=83.5, N_phi=48
Adaptive mode: target relative change = 0.01
  N=64  (+64 new), M11(180)=7.09, dM11=100%, Csca=92.9, dCsca=100%
  N=128 (+128 new), M11(180)=6.69, dM11=5.6%, Csca=93.0, dCsca=0.12%
  N=256 (+256 new), M11(180)=6.64, dM11=0.8%, Csca=92.9, dCsca=0.08%
Converged at N=256 (both C_sca and M11)
```

Convergence messages:

- `Converged (both C_sca and M11)` — best: both criteria met
- `Converged (C_sca ok, M11 within X%)` — relaxed: M11 within $5 \times \varepsilon$
- `Converged (C_sca stable, M11 still X%)` — C_sca $< \varepsilon/10$ after 4 iterations
- `WARNING: Max orientations reached` — did not converge, result approximate

4.2.2 Sobol mode output

```
Sobol: 1024 orientations, beta_sym=30 deg, gamma_sym=180 deg
Phase 1 (tracing + preprocessing): 0.46 s
Phase 2 (diffraction, OpenMP): 13.25 s
Total: 13.71 s
```

4.2.3 Scattering efficiency (always printed)

```
===== SCATTERING EFFICIENCY =====
C_sca = 160.0398
A_proj (incoming energy) = 93.0208
Q_sca = C_sca / A_proj = 1.7205
================================
```

If $Q_{\text{sca}} > 2.5$: `WARNING: Q_sca > 2.  PO overestimates at this size.`

4.3 Console output (stderr)

4.3.1 Memory diagnostic

```
Memory: 25219 MB available, chunk=4096 orientations (1 chunks)
Adaptive: max orientations = 32768 (25219 MB available)
```

4.3.2 Timing breakdown (at program exit)

```
===== HandlerPO TIMING BREAKDOWN =====
HandleBeams calls: 448
HandleBeams total:    69.255 s
  Beam loop total:    66.039 s
  AddToMueller:       2.606 s
  Unaccounted:        63.433 s
=====================================
```

"Unaccounted" = the optimized inline loop (batched sincos + edge loop + rotate_jones). Not instrumented to avoid timing overhead.

5 Build

5.1 Intel / AVX-512

```
bash build.sh            # -> bin/mbs_po
```

`-O3 -march=native -mavx512f -mavx512dq -fopenmp`. Requires GCC ≥ 9.

5.2 AMD EPYC 7H12 (Zen 2, SP3)

```
bash build_epyc.sh       # -> bin/mbs_po_epyc
bash build_epyc_clang.sh # -> bin/mbs_po_epyc_clang
```

`-march=znver2`. No AVX-512; `fast_sincos_8x` falls back to 2×`fast_sincos_4x`.

NUMA: `OMP_PROC_BIND=close OMP_PLACES=cores OMP_NUM_THREADS=64`

5.3 AMD Zen 4 (Ryzen 7000, EPYC Genoa)

```
bash build_zen4.sh       # -> bin/mbs_po_zen4
```

`-march=znver4 -mavx512f -mavx512dq -mavx512vl`.

5.4 Generic AVX2

Remove `-mavx512*` from `build.sh`. Code auto-detects via `#ifdef __AVX512F__`.

6 Performance

6.1 Single-thread benchmarks

Hex D=H=10 μm, 128 Sobol, 48 ϕ × 181 θ, n=6:

Version	Phase 2	vs Original
Original (before optimizations)	∼320 s	1×
Optimized + batched sincos	4.6 s	∼**70**×
EPYC build (AVX2 only)	4.7 s	∼68×

6.2 Key optimizations

1. **PrecomputeEdgeData**: slopes, intercepts computed once per beam (not per direction).
2. **Theta-coefficients**: $A(\theta) = \sin\theta \cdot a_s + \cos\theta \cdot a_c + a_0$ (2 FMA).
3. **Batched sincos**: all vertex phases for all θ pre-computed before the θ-loop. AVX-512 `fast_sincos_8x` or AVX2 2×`fast_sincos_4x`.

4. **Polynomial sincos**: Cody–Waite range reduction + Chebyshev polynomial (∼14 decimal digits).
5. **Inline hot loop**: no heap allocation, stack-only `complex` arithmetic.
6. **OpenMP**: thread-local Mueller accumulators, `schedule(dynamic,1)`.
7. **Sobol + symmetry**: native C++ Sobol generator with particle symmetry reduction.

6.3 OpenMP scaling

Threads	Phase 2	Speedup
1	4.6 s	1.0×
4	1.7 s	2.7×
12 (SMT)	∼0.9 s	∼5×
64 (EPYC)	∼0.2 s*	∼23×*

*Estimated; use ≥512 orientations for 64 cores.

7 Bugfixes

7.1 Forward-direction Fresnel sign (2026-03-22)

Bug: sign error in the forward-direction special case ($A{\approx}0, B{\approx}0$) of the Kirchhoff integral. The edge-sum converges to $-F_{\text{inv}} \cdot \text{area}$ but the shortcut used $+F_{\text{inv}} \cdot \text{area}$.

Symptom: $M_{11}(0°)$ anomalously small in coherent mode for small particles. Example (hex $D{=}H{=}3\,\mu$m): $M_{11}(0)/M_{11}(0.1°) = 0.019$ (should be ≈ 1.03).

Fix: negate the forward-direction formula in 6 locations:

- `HandlerPO.cpp`: lines 970, 1333, 1889
- `Handler.cpp`: lines 255, 473, 642

Incoherent mode unaffected ($|J|^2$ is sign-independent). Q_{sca} changed by ∼0.06%. See `BUGFIX_forward_direct` for full details.

8 Known Limitations

1. $Q_{\text{sca}} > 2$ **at large** x: PO does not enforce the optical theorem.
2. M_{33}, M_{34}, M_{44}: RotateJones vs Karczewski give different off-diagonal elements (same M_{11}).
3. **Small particles** ($x < 20$): PO not valid. Use ADDA or T-matrix.
4. **Backscattering convergence**: $M_{11}(180°)$ needs 10–100× more orientations.
5. **LTO**: `-flto` causes ∼20% regression. Do not use.

9 File Listing

File	Description
`build.sh`	Intel/AVX-512 build
`build_epyc.sh`	AMD EPYC 7H12 (Zen 2) build
`build_epyc_clang.sh`	EPYC + Clang/AOCC build
`build_zen4.sh`	AMD Zen 4 (Ryzen 7000/Genoa) build
`Makefile`	Legacy Makefile (Intel)
`Makefile.epyc`	Makefile for EPYC
`tests/run_tests.sh`	Regression test suite
`MANUAL.md`	This manual (Markdown source)
`BUGFIX_forward_direction_sign.md`	Bugfix documentation
`docs/bugfix_15x_mismatch.pdf`	15× mismatch investigation

10 Bugfix: 15× Mismatch (old vs new binary)

Comparison of `-fixed 45 30` between old and new binaries showed a 15.5× difference in M_{11}.

10.1 Root causes

1. **Old `-fixed` never rotated the particle.** `ScatteringConvex::ScatterLight()` has `m_particle->Rotate` commented out, and old `TracerPO::TraceFixed` did not call it either. Result: `-fixed 45 30` always gave the default orientation. New code correctly adds `m_particle->Rotate(b, g, 0)` in `TraceFixed`.
2. **Uninitialized `d_param` in particle constructors.** `Point3f()` does not zero `coordinates[3]`. No constructor called `SetDParams()` after `Reset()`. Fixed: `SetDParams()` added after `Reset()` in all 8 particle constructors.
3. **Mixed `fast_exp_im`/`exp_im` in fallback paths.** `DiffractIncline`, `DiffractInclineFast`, `ComputeFnJones`, and `ApplyDiffraction*` had inconsistent precision. Fixed: all fallback paths use `exp_im` (glibc sincos).

10.2 Validation

Test	Max diff	Mode	Notes
`-fixed 0 0 -incoh`	exact	incoh	per-beam $\|J\|^2$ identical
`-random 64 43 -incoh`	exact	incoh	per-beam physics identical
`-random 32 22 -coh_orient`	10^{-7}	coh	machine precision
`-random 64 43` (default)	0.007%	incoh avg	coh vs incoh averaging
HandleBeams inline vs fallback	identical		SIMD optimizations verified

Speedup: 55× (23 min → 25 sec at 2860 orientations, 4 threads).

d_param note: `d_param` is the plane equation parameter ($ax + by + cz + d = 0$) stored in `Point3f::coordinates[3]`. It was uninitialized before `Rotate()` call. `SetDParams()` now added to all particle constructors. This changes coherent single-orientation M_{11} by ∼7% (phase-sensitive), but incoherent $|J|^2$ is identical. After orientation averaging the effect is $< 0.01\%$.

Impact: Only `-fixed` mode affected. Orientation averaging modes (`-random`, `-sobol`, `-adaptive`, `-auto`) always call `Rotate()` via `TracerPOTotal`, so d_param is always initialized correctly.

See `docs/bugfix_15x_mismatch.pdf` and `tests/reference_test/` for full reports and plots.

11 New Features (March 2026)

11.1 Adaptive theta grid (`-auto_tgrid EPS`)

Replaces the static zone-based theta grid with recursive bisection. For each interval $[\theta_i, \theta_{i+1}]$: computes M_{11} at the midpoint, compares with linear interpolation. If relative error > EPS, splits the interval.

- Minimum step: $\lambda/(8D)$ (quarter-Nyquist)
- Maximum depth: 8 levels
- Probe orientations: div16 physics estimate (same as adaptive start)
- Result: 2–3× fewer theta points than static grid

Usage: `-auto_tgrid 0.05` (5% tolerance). Works with `-sobol`, `-random`, `-auto`.

11.2 Flag priorities

Multiple flags can set the same parameter. Resolution order (highest priority first):

Theta grid (N_θ and theta values)

1. `-tgrid FILE` — explicit non-uniform grid from file
2. `-grid T1 T2 Nphi Ntheta` — uniform grid from command line
3. `-auto_tgrid EPS` — adaptive bisection (probes M_{11})
4. `-auto` — implies `-auto_tgrid 0.05` (bisection)
5. `-oldauto DIV` — static zone grid from `ApplyAutoThetaGrid`
6. Default: placeholder (1 theta point)

Phi grid (N_φ)

1. `-nphi N` — explicit override (highest priority)
2. `-grid T1 T2 Nphi Ntheta` — `Nphi` is the 3rd argument
3. `-auto_phi` — formula $N_\varphi = \min(360,\ x/5 + 48)$
4. `-auto` — implies `-auto_phi`
5. `-oldauto DIV` — same formula as `-auto_phi`
6. Default: 1 (useless without override)

Orientation method

Mutually exclusive (first matched wins):

1. `-fixed BETA GAMMA` — single orientation
2. `-oldauto DIV` — physics-based $\beta \times \gamma$ grid
3. `-random Nb Ng` — regular grid
4. `-montecarlo N` — random uniform
5. `-orientfile FILE` — from file
6. `-auto EPS` / `-autofull EPS` — adaptive Sobol
7. `-adaptive EPS` — adaptive Sobol (manual grid)
8. `-sobol N` — fixed-count Sobol

Beam cutoff

1. `-beam_cutoff EPS` — explicit threshold
2. `-auto`/`-adaptive` — uses their EPS as cutoff
3. Default: 0 (no cutoff)

Coherent/incoherent

1. Default: coherent (Jones summed across beams per orientation)
2. `-incoh`: incoherent (Mueller per beam)
3. `-coh_orient`: coherent across ALL orientations (legacy)

Note: `-grid T1 T2 Nphi Ntheta` — the 3rd value is N_φ (not N_θ!). This is the original MBS convention. Use `-nphi` to avoid confusion.

11.3 Multi-size (`-multigrid Dmin Dmax N`)

Generates N sizes from $D_{\min}$ to $D_{\max}$ in logarithmic scale. For each size: rescales particle via `Resize()`, runs full `TraceFromSobol` (same optimized path as standalone). Results bit-exact with separate runs.

```
# 10 sizes from D=50 to D=500, reference particle D=500
bin/mbs_po --po --sobol 1024 --multigrid 50 500 10 \
    -p 1 500 62.5 -w 0.532 --ri 1.31 0 -n 8 \
    --grid 0 180 360 180 --close -o results
```

Output: `results_x{N}.dat` for each size parameter $x = \pi D/\lambda$.

11.4 Per-beta save (`-save_betas`)

With `-random`: writes per-beta Mueller contribution and cumulative Mueller to `_betas/` subfolder. Useful for convergence analysis and crash recovery.

11.5 Parallel Phase 1 (tracing)

Beam tracing (Phase 1) is now parallelized with OpenMP. Each thread gets its own copy of `Particle` + `Scattering` via `CloneFor()`. Phase 1 speedup: 5.4× on 12 threads.

11.6 Additional bugfixes

- `-montecarlo` missing handler config (isCoh, m_wave, shadowOff)
- `DiffractControlPoints`: wrong M11 formula (missing d01, d10 terms)
- `-karczewski` ignored in fast diffraction path
- `rdtsc` inline asm incorrect on x86-64
- `malloc` leak in `Particle::SetFromFile`
- OOM in `TraceAdaptive`: sub-chunked to 4096 max
- `-grid` conflicts with auto theta/phi resolved
- `-oldauto` N rounding: ceil instead of ODD (smoother size scans)
- `-point` crash from uninitialized handler pointer
- MPI missing `}` in `TraceAdaptive` convergence block

12 Performance: MBS-fast vs MBS-raw (original)

All benchmarks: hex column $L = 199.5$, $D = 24.9375$ µm, $\lambda = 0.532$, $m = 1.31$.

The original MBS-raw code is **single-threaded**, uses standard `sin`/`cos` (glibc), computes the diffraction integral with full `exp_im` per edge, and has no beam filtering or adaptive grids.

12.1 Architecture comparison

Component	MBS-raw (original)	MBS-fast
Phase 1 (tracing)	1 thread, sequential	OpenMP, `CloneFor()` per thread
Phase 2 (diffraction)	1 thread, `exp_im` per edge	OpenMP, batched `fast_sincos` (AVX)
Edge integral	`ChangeCoordinateSystem` per vertex	Precomputed edges + slopes
A,B coefficients	Recomputed per (beam, θ, φ)	Theta-coefficients: hoist per φ
Vertex phases	2 × `exp_im` per edge per θ	1 × `fast_sincos` per vertex, batched
Beam filtering	None (all beams computed)	Two-threshold cutoff (skip 45–80%)
θ grid	Uniform or hand-tuned zones	Adaptive bisection (`-auto_tgrid`)
φ grid	Manual `-grid`	Auto: $N_\varphi = x/5 + 48$
Orientations	Regular $\beta \times \gamma$ grid	Sobol quasi-random + adaptive
Memory	Full arrays, no chunking	Chunked streaming (4096 max)

12.2 Speedup breakdown

Each optimization contributes a multiplicative factor. Combined effect depends on which phase dominates (Phase 2 for most cases).

Optimization	Where	Factor	Notes
Phase 2 (diffraction) — typically 90–95% of total time:			
OpenMP (thread-local Mueller)	Phase 2	$N_{\text{cores}}\times$	12 cores → 10–11×
Batched sincos (AVX-512 8-wide)	Phase 2	∼2×	polynomial approx, SIMD
Precomputed edge data	Phase 2	∼1.5×	slopes, intercepts cached
Theta-coefficients	Phase 2	∼1.3×	A,B decomposed per φ
Beam cutoff	Phase 1+2	∼2–4×	skip weak beams
Adaptive theta (bisection)	Grid	∼2–3×	fewer θ points
Phase 1 (tracing) — typically 5–10% of total time:			
OpenMP (`CloneFor`)	Phase 1	$N_{\text{cores}}\times$	12 cores → 5.4×
Adaptive mode:			
DiffractControlPoints	Convergence	∼100×	4 angles, not full grid
Incremental Sobol	Convergence	∼3.5×	reuse previous orientations
Physics-based start (div16)	Convergence	∼2×	skip initial low-N iterations

Combined estimate for 12 cores:

Phase 2: 10× (OMP) × 2× (AVX) × 1.5× (edges) × 1.3× (theta-coeff) × 2× (beam cutoff) ≈ **78**×. With adaptive theta (2×): ≈ **156**×.

12.3 Measured speedup

Test	Params	Old (1 thr)	New	Speedup	Accuracy
Forward 0–25°	$n=4$, 64×43, 4 thr	17 min	16 sec	**63**×	0.007%
Forward 0–25°	$n=4$, 64×43, 12 thr	17 min	11.5 sec	**89**×	0.007%
Backward 160–180°	$n=11$, 64×43, 4 thr	47 min	27 sec	**105**×	0.8%
Backward 160–180°	$n=11$, 64×43, 12 thr	47 min	17 sec	**168**×	0.8%
`-coh_orient`	32×22, 4 thr	30 sec	0.5 sec	**60**×	10^{-7}
Phase 1 only	$n=8$, 8192, 12 thr	5.2 sec	0.96 sec	**5.4**×	exact

On 128-core EPYC cluster with MPI: ∼1000×+ expected.

12.4 Accuracy

- `-coh_orient`: machine precision (10^{-7}) match with original
- `-incoh`: exact per-beam $|J|^2$ — physics identical
- Default mode: $< 0.01\%$ at convergence (coh vs incoh averaging)
- Forward peak ($\theta=0$): 0.39% from `d_param` fix (correct value)
- All optimizations verified: inline path = fallback path (bit-exact)